

Introductory Econometrics with Recitation: TA Session Week 11

Danny Po-Hsien Kang (康柏賢)

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Today's agenda

- 1 Regular Expressions
 - `sub()`, `gsub()`
 - Regular Expressions

- 2 Regressions
 - Polynomial Terms
 - `geom_smooth()`
 - Interaction Terms
 - `predict()`

Today's Dataset

- Today we will use the following CSV file from the OpenIntro Website:
 - `acs12.csv`: Results from the US Census American Community Survey, 2012.
- Please import the following datasets into RStudio.

```
acs <- read.csv("data/acs12.csv")

acs <- acs %>%
  na.omit() %>%
  mutate(univ = ifelse(edu == "hs or lower", 0, 1)) %>%
  filter(income != 0)
```

String Replacement: sub() and gsub()

- We use sub() and gsub() to replace text inside a string.
- The difference is:
 - sub(): replace the **first match** only
 - gsub(): replace **all matches**
- **Syntax:**

```
sub(pattern, replacement, x)  
gsub(pattern, replacement, x)
```

- pattern: the text or regular expression you want to find
- replacement: the new text
- x: the string or character vector

String Replacement: sub() and gsub()

- **Example:**

```
text <- "apple apple apple"

sub("apple", "pear", text)
gsub("apple", "pear", text)

gsub("-", "/", c("2025-04-22", "2025-04-29"))
```

- **Output:**

```
[1] "pear apple apple"
[1] "pear pear pear"

[1] "2025/04/22" "2025/04/29"
```

Regular Expressions: Basic

- A **regular expression** (regex) is a pattern used to match text.
- We often use regex with: `sub()`, `gsub()`, `grep1()`
- Regex lets us match a rule or pattern.
- **Common regex symbols:**

<code>.</code>	any single character
<code>^</code>	start of a string
<code>\$</code>	end of a string
<code>[abc]</code>	one of a, b, c
<code>[0-9]</code>	one digit from 0 to 9
<code>+</code>	one or more times
<code>*</code>	zero or more times

- These symbols are very useful in cleaning text data.

Regular Expressions: ^ and \$

- **Example:**

```
words <- c("cat", "catalog", "mycat")  
  
grepl("^cat", words)  
grepl("cat$", words)
```

- **Output:**

```
[1] TRUE TRUE FALSE  
[1] TRUE FALSE TRUE
```

- ^cat means the string starts with "cat".
- cat\$ means the string ends with "cat".

Regular Expressions: [0-9]

- **Example:**

```
text <- c("room12", "apple", "A7")  
  
grepl("[0-9]", text)  
gsub("[0-9]", "", text)
```

- **Output:**

```
[1] TRUE FALSE TRUE  
[1] "room" "apple" "A"
```

- [0-9] matches any digit from 0 to 9.

Regular Expressions: .

- **Example:**

```
strings <- c("cat", "cot", "cut", "coat")  
  
grepl("c.t", strings)
```

- **Output:**

```
[1]  TRUE  TRUE  TRUE FALSE
```

- . means any single character.
- So "c.t" matches "cat", "cot", and "cut".

Regular Expressions: * and +

• Meaning:

<pre>* zero or more times + one or more times</pre>
--

• `ab*` means:

- "a"
- "ab"
- "abb"
- "abbb", ...

• `ab+` means:

- "ab"
- "abb"
- "abbb", ...

Regular Expressions: * and +

- **Example:**

```
strings <- c("a", "ab", "abb", "abbb", "ac")  
  
grepl("ab*", strings)  
grepl("ab+", strings)
```

- **Output:**

```
[1] TRUE TRUE TRUE TRUE TRUE  
[1] FALSE TRUE TRUE TRUE FALSE
```

- `ab*` matches "a" first, and also matches the first part of "ac".
- `ab+` requires at least one "b", so it does not match "a" or "ac".

Regular Expressions: Combining Symbols

- We can combine regex symbols to match more flexible patterns.
- `.*` is often used to mean **“anything in between”**.
- **Example:**

```
emails <- c("alice@gmail.com", "bob@ntu.edu.tw")  
sub("@.*", "", emails)
```

- **Output:**

```
[1] "alice" "bob"
```

Nonlinear Regression: Polynomial Terms

- **We consider the following model:**

$$\text{income}_i = \beta_0 + \beta_1 \text{age}_i + \beta_2 \text{age}_i^2 + \varepsilon_i$$

- The quadratic term is included in the model.
- How do we create the quadratic term in R?

```
acs <- acs %>%  
  mutate(age2 = age * age)
```

- We can also use a built-in function to create polynomial terms when estimating a model.

Nonlinear Regression: Polynomial Terms

• Syntax:

```
lm_model_poly <- lm_robust(  
  income ~ age + I(age^2),  
  data = acs  
)
```

• Output:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-67741.602	19928.832	-3.399	0.000
age	4853.192	1163.161	4.172	0.000
I(age^2)	-47.235	14.972	-3.155	0.002

Visualizing Regression: `geom_smooth()`

- `geom_smooth()` adds a fitted regression line to a scatter plot.

- **Syntax:**

```
ggplot(acs, aes(x = age, y = income, color = ...)) +  
  geom_point() +  
  geom_smooth(method = "lm", formula = ..., se = ...)
```

- `method = "lm"` tells R to fit a linear regression model.
 - default: `formula = y ~ x`
- `geom_smooth()` also plots the confidence interval.
 - `se = FALSE` removes the shaded confidence interval.
- Adding `color = ...` inside `aes()` groups the data by that variable and plots separate colors.
 - For example, `aes(x = age, y = income, color = citizen)`

Visualizing Regression: Polynomial Fit

- We can visualize a polynomial regression model.
- **Example:**

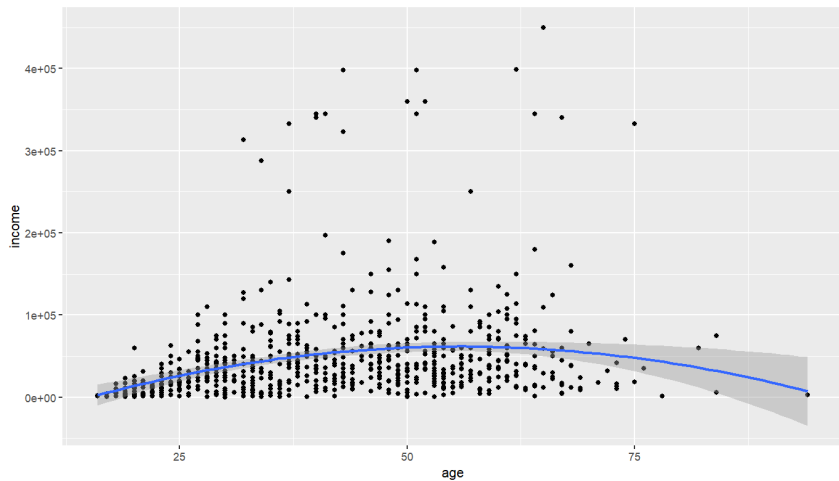
```
ggplot(acs, aes(x = age, y = income)) +  
  geom_point() +  
  geom_smooth(  
    method = "lm",  
    formula = y ~ x + I(x^2)  
  )
```

- This corresponds to the quadratic regression model:

$$\text{income}_i = \beta_0 + \beta_1 \text{age}_i + \beta_2 \text{age}_i^2 + \varepsilon_i$$

- We can use `aes(color = ...)` when there are multiple groups.

Visualizing Regression: Polynomial Fit



Nonlinear Regression: Interaction Terms

- **Syntax 1:**

```
lm_model_interaction <- lm_robust(  
  income ~ age * univ,  
  data = acs  
)
```

- **Syntax 2:**

```
lm_model_interaction <- lm_robust(  
  income ~ age + univ + age:univ,  
  data = acs  
)
```

- These two specifications are equivalent.

Nonlinear Regression: Interaction Terms

- **Output:**

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	10997.671	3589.542	3.064	0.002
age	505.137	82.779	6.102	0.000
univ	-2276.693	14040.242	-0.162	0.871
age:univ	913.643	357.976	2.552	0.011

Prediction: predict()

- **Syntax:**

```
predict(model, newdata = data_frame)
```

- First, we fit a model of income on hours worked.

```
model <- lm_robust(income ~ hrs_work, data = acs)
```

Prediction: predict()

- Model output:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-16905.310	7511.361	-2.251	0.025
hrs_work	1641.663	209.689	7.829	0.000

- Suppose there are two workers with different working hours:

```
# Initial working hour is 40hrs
low_hr <- data.frame(hrs_work = 40)

# Now increase by 2hrs to 42hrs
high_hr <- data.frame(hrs_work = 42)
```

- Income is predicted to increase by $2 \times 1641.663 = 3283.326$ dollars.

```
predict(model, high_hr) - predict(model, low_hr)
## 3283.326
```